

Quarterly Earnings Intelligence Platform

An Agentic, Multi-Source Financial Document Q&A and Engineering System for Indian Capital Markets

A consolidated architectural definition blueprint merging intelligent multi-agent LangGraph orchestration with production-ready cloud deployment, Streamlit interface dynamics, and enterprise-grade multi-source ingestion pipelines.

Theme Assignment:	Theme 12 — LangGraph Agents, RAG Pipelines & NSE/BSE Intel
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Target Stack:	Streamlit, LangGraph, LangChain, ChromaDB, Kubernetes, Helm, ArgoCD
Classification:	Commercial Confidential / Product Blueprint
Date:	May 2026

1. Executive Summary & Market Context

Modern capital markets generate vast quantities of regulatory filings, comprehensive financial disclosures, and critical investor relations transcripts each quarter. Within the Indian ecosystem, the National Stock Exchange (NSE) and Bombay Stock Exchange (BSE) list over 5,000 corporate entities. Each entity issues mandated statutory reports on disjointed timelines throughout the earnings cycle, presenting an immense manual collation overhead.

5,000+

LISTED COMPANIES TRACKED

4-6 Hours

DAILY ANALYST TIME SAVED

\$61.3B

GLOBAL FINTECH AI MARKET
(2031)

Institutional equity research teams, fund managers, and venture capital associates spend roughly 4 to 6 hours per day browsing the BSE XBRL portal, filtering through the NSE corporate announcements terminal, and manually sorting local IR repositories. Current commercial options remain highly restrictive: general-purpose LLM interfaces lack multi-tenant workflow continuity and auto-refresh mechanisms, while legacy database solutions focus on structured metrics without interpreting qualitative commentary or tracking cross-company narrative evolutions.

The **Quarterly Earnings Intelligence Platform** resolves these foundational inefficiencies. By establishing an autonomous, zero-touch ingestion pipeline combined with a multi-agent orchestration architecture, it abstracts the discovery, downloading, chunking, indexing, and qualitative analysis of Indian corporate disclosures into a seamless conversational knowledge system.

2. Consolidated Functional Requirements & UX Specification

Transitioning the initial technical solution into a robust enterprise portal requires integrating explicit UX workflows with background ingestion constraints. The user-facing layers are designed around a secure, containerized Streamlit dashboard mapped directly into automated enterprise storage clusters.

2.1 Identity Management & Cloud Sign-On

- **Unified Authentication Window:** System access is strictly gated behind enterprise Single Sign-On (SSO) and Multi-Factor Authentication (MFA) mechanisms.
- **Cloud Credential Integration:** Session tokens and security context map directly to cloud environment access parameters (AWS/GCP), establishing secure down-scoped database and bucket access per user tenant.

2.2 Active Corporate On-boarding & Coverage Discovery

- **On-boarded Company Workspace:** A dynamic administrative panel displaying all companies currently being tracked by the extraction engines.
- **AutoComplete Discovery Tool:** An interactive drop-down interface featuring typeahead autocomplete. It fetches comprehensive company master records directly from the exchange registry database.
- **Workflow Trigger Initialization:** Users can manually initiate the automated multi-agent ingestion workflow for newly selected entities to instantly pull historical and current operational data.

2.3 Multi-Source Data Scope & Source Mapping

The platform relies on an expanded ingestion surface area structured via an abstract **NSE MCP Server** architecture. Discovered files are ingested and tagged under specific thematic taxonomies:

Source Origin	Technical Mapping Channel	Target Document Taxonomies
NSE Corporate Portal	Official Filings & Announcements API	1. Financial Results, 2. Quarterly/Annual Reports
Corporate IR Pages	Playwright Headless Browser Scrapers	2. Quarterly Reports, 3. Corporate Announcements
Exchange Registrars	NSE/BSE Corporate Actions Feed	4. Shareholding Patterns, 5. Corporate Governance
Market Data Feeds	Structured Pricing & Consensus Providers	6. Board Meetings, 7. Corporate Actions
Credit & Industry Feeds	Rating Agencies & Regulatory Transcripts	3. Statutory Announcements & Risk Profiles

2.4 Synchronous Lifecycle Management & File Ingestion

- **Dynamic Schedulers:** Users can configure precise polling schedules (e.g., 15-minute intervals for active earnings days or daily cycles for off-peak periods).
- **Ad-Hoc Ingestion Overrides:** Analysts can bypass scheduled intervals via direct manual file uploads (PDF/Image formats) or target specific URLs for live web scraping.
- **Internal Auditing History:** A persistent logging panel tracks ingestion attempts, payload sizes, processing states, and vector indexing history.

2.5 The Automated Intelligence Workspace

- **Multi-Dimensional Corporate Summaries:** The system surfaces multi-faceted entity profiles covering operational business metrics, financial fundamentals, and technical price-action indicators.
- **Exchange Correspondence Tracker:** Live capture of clarifications, inquiries, and regulatory responses between listed corporations and stock exchanges.
- **Categorized KPI Matrix:** Segmented extractions isolating growth vectors, balance sheet liquidity, operational margins, and structural risk flags.
- **Conversational Chat Interface:** An integrated Streamlit conversation portal connected directly to the underlying retrieval-augmented generation graph.

3. Production Technical Stack

The underlying technology stack has been refined to shift away from proprietary, black-box third-party tools toward open-source, highly auditable components that facilitate isolated, on-premise or secure cloud container deployments.

Layer	Component Selection	Operational Execution Role
User Interface	Streamlit TailwindCSS	Rapid reactive UI generation for dashboards and chatbot components.
Agent Core	LangGraph LangChain	Stateful multi-agent cyclic graph compilation and validation.
Document Splitter	RecursiveCharacterTextSplitter	Context-aware, token-bounded document splitting.
Vector Engine	ChromaDB	Local, lightning-fast vector space for context querying.
Relational Cache	In-Memory File-Based DB	Lightweight local key-value and metadata state caching.
Scraping Framework	Playwright BeautifulSoup4	Dynamic SPA rendering handling and deep DOM parsing.
Sentiment Classifier	FinBERT Model NLP Market Analyser	Financial tone categorization based on open-source repositories.
Orchestration	Kubernetes Helm Charts	Scalable container execution and microservices provisioning.
CI/CD GitOps	Jenkins ArgoCD GitHub	Continuous integration pipelines and declarative GitOps delivery.

4. Multi-Agent System Architecture & Workflow

The backbone of the intelligence platform is a stateful Directed Acyclic Graph (DAG) built via **LangGraph**. This architecture ensures each component functions as an isolated agent equipped with strict specialized tools, maintaining deterministic execution paths across asynchronous ingestion pipelines.

The Stateful Ingestion & Analytical Lifecycle Blueprint:

[Scraping Agent] —(SHA-256 Check)—> [Processing Agent] —(Recursive Split)—> [ChromaDB Vector Store]

Agent]

└─(Parallel Nodes)—> [KPI Agent] & [Sentiment

└─(State Update)—> [Comparison Agent] —>

[Memo Node]

4.1 Specialized LangGraph Agent Node Definitions

- 1. **Automated Scraping Agent:** Uses headless Playwright sessions to track the NSE corporate announcements registry and company investor relations modules. It extracts raw document binary arrays, computes unique SHA-256 hashes to prevent processing duplicates, and updates the local metadata state cache.
- 2. **Document Processing Agent:** Accepts raw document streams. It extracts structural textual tables and paragraphs, using token-bounded splitting algorithms to group blocks into clean, 512-token chunks with a 64-token overlap.
- 3. **KPI Extraction Agent:** Applies structured schema parsing constraints onto extracted text tables to extract critical parameters (Revenue, EBITDA, PAT, EPS, and Operating Margins), passing typed dictionaries to internal storage.
- 4. **Market Sentiment Analysis Agent:** Processes transcript arrays through an integrated FinBERT model wrapper, scoring tone distribution (bullish, neutral, cautious) and identifying key phrases or management posture shifts.
- 5. **Cross-Company Comparison Agent:** Queries peer groups across matching temporal vectors, constructing differential matrices that detail quarter-over-quarter metrics and narrative shifts.
- 6. **RAG-Driven Retrieval Agent:** Handles conversational chat requests. It executes semantic matching routines across ChromaDB embeddings, returning ranked context blocks complete with verifiable source citations.
- 7. **Report Generation Agent:** Pulls extracted KPI structures and active risk/sentiment vectors to compile structured investment memos and automated operational update sheets.

5. Expected Operational Outcomes & Strategic Business Impact

Deploying the platform drastically reduces analytical response times, shifting researcher workflows from manual data collection to immediate strategic evaluation.

Workflow Metric Matrix	Manual Baseline	Automated Platform State	Productivity Gains
Regulatory Ingestion & Discovery	4 – 6 Hours / Day	< 10 Minutes (Alert Notification)	~95% Time Reduction
Financial Summary Generation	45 – 90 Minutes / Doc	< 2 Minutes (Automated Memo)	~97% Processing Delta
Cross-Company Sector Synthesis	2 – 3 Hours / Excel	Instantaneous Real-time Query	Eliminates Manual Copying
Earnings Transcript Evaluation	60 – 90 Minutes / Call	5 Minute FinBERT Sentiment Split	~92% Focus Improvement
Filing Omission Vulnerability	Variable / High Risk	Near-Zero (Deterministic Polling)	Risk Completely Mitigated

For an institutional group maintaining 10 active buy-side or sell-side financial analysts, this automated intervention reclaims 300 to 500 working hours every month. The resulting increase in analysis capacity allows institutions to

scale their coverage universe without expanding headcount. This efficiency transforms research operations from simple historical document collators into highly accurate forward-looking market monitors.